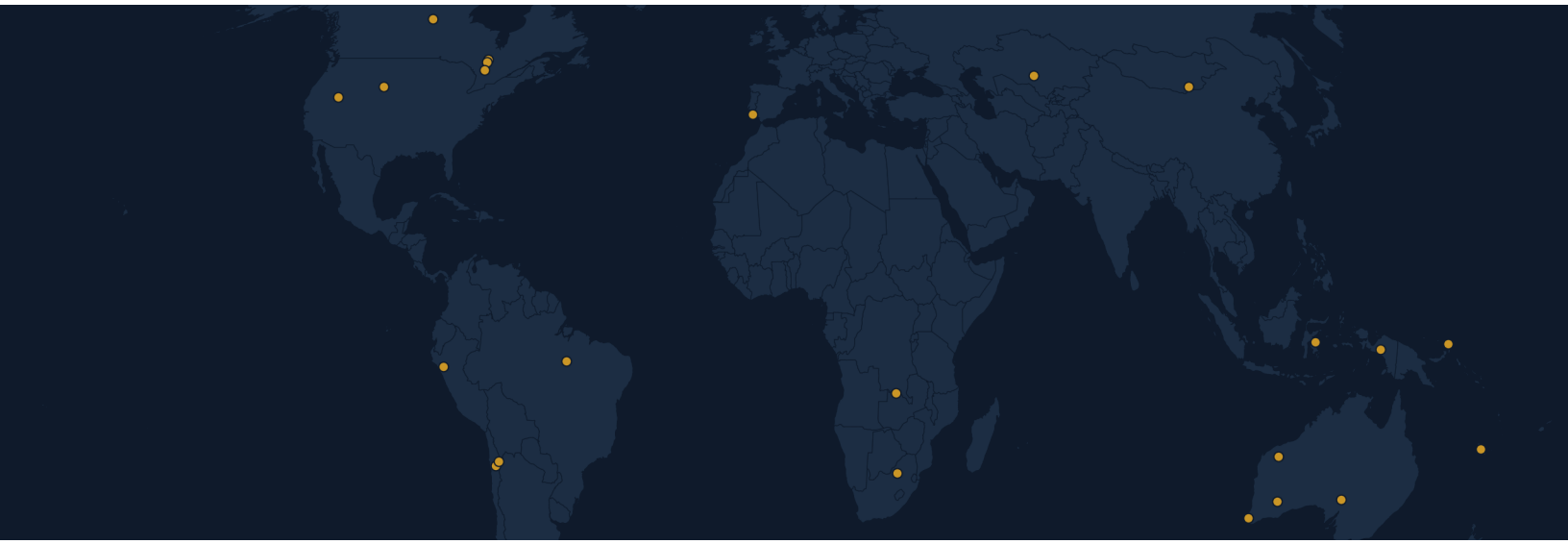
METALS & MINING MADE SIMPLE

VALUE DRIVERS FOR INVESTMENT PROFESSIONALS

METALS & MINING RESEARCH

Basel Al-Ghousein, Analyst



Educational research. Market data as of October 27, 2026. Important disclosures appear at the end of this report.

NORTHWALL RESEARCH

Metals & Mining Made Simple (First Edition)

Value drivers for investment professionals, from rock in the ground to a share price.

This primer explains what drives value in metals and mining equities. It follows the mine value chain first (geology, resources, mining and processing), then the major commodities, then costs, fiscal terms and valuation. Two appendices walk through a sample drill-results press release and a glossary of over 100 terms. Market figures are as of October 2026 unless stated; worked examples use round, hypothetical numbers.

Throughout, we use Northwall Research's own coverage to show how the concepts appear in practice: Alamos Gold for operating leverage and cost trajectory, and Franco-Nevada for the royalty model and premium valuation.

Metals & Mining Research

Basel Al-Ghousein (Analyst), Northwall Research. Level II candidate for the CFA designation.

Company	Ticker	Sub-sector	Rating	Target
Alamos Gold Inc.	NYSE/TSX: AGI	Gold producers	Buy	US$39.00
Franco-Nevada Corporation	NYSE/TSX: FNV	Gold royalties and streaming	Hold	US$250.00

Ratings and targets as published October 27, 2026. Research and live coverage: kosherite.github.io/Northwall-Research

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Abbreviations and Units

oz	Troy ounce (31.1035 g)	**LCE**	Lithium carbonate equivalent
koz / Moz	Thousand / million troy ounces	**U_3O_8**	Uranium oxide concentrate
g/t	Grams per tonne (precious metals grade)	**AISC / AIC**	All-in sustaining cost / all-in cost
% Cu, % Ni	Percent metal by weight (base metals grade)	**C1**	Cash cost (Wood Mackenzie definition)
lb	Pound (copper, nickel, zinc, uranium prices)	**NSR**	Net smelter return
ppm	Parts per million (1 ppm = 1 g/t)	**TC/RC**	Treatment and refining charges
AuEq / CuEq	Gold / copper equivalent grade	**NAV / P/NAV**	Net asset value / price to NAV
t / kt / Mt	Metric tonne / thousand / million tonnes	**LOM**	Life of mine
tpd / Mtpa	Tonnes per day / million tonnes per annum	**PEA / PFS / FS**	Preliminary economic assessment / pre-feasibility / feasibility study
dmt / wmt	Dry / wet metric tonne	**GEO**	Gold-equivalent ounce

Conversions used throughout: 1 t = 32,150.7 oz; 1% Cu = 22.046 lb Cu per tonne of ore; 1 g/t Au = 0.03215 oz per tonne of ore.

Metals & Mining Defined

A mining equity is a leveraged claim on metal in the ground. Its value depends on how much metal there is, at what grade, at what cost to extract, in which jurisdiction and at what price.

Unlike oil and gas, there is no single "barrel." The first question for any mining stock is which commodity drives its cash flow.

Unlike oil and gas, there is no single "barrel" in mining. Each metal has its own demand drivers, price benchmark and processing route. The first question for any mining stock is therefore which commodity drives its cash flow, and the second is where the company sits on the value chain.

Commodity Families

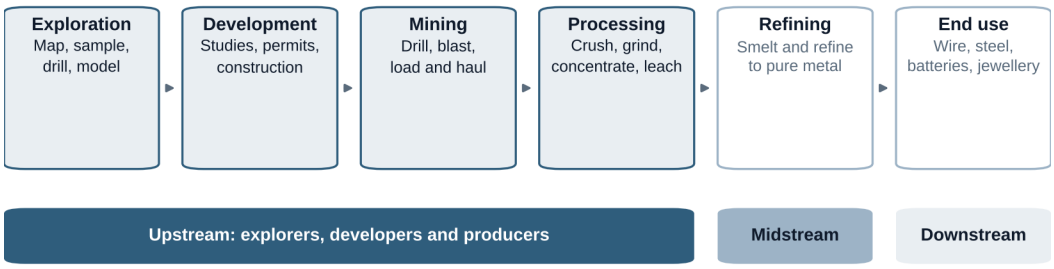
Family	Examples	Main demand drivers	Where the price is set
Precious metals	Gold, silver	Investment (ETFs, bars, coins), central-bank buying, jewellery; silver also industrial (solar, electronics)	LBMA (London), COMEX (New York)
Platinum group metals	Platinum, palladium, rhodium	Autocatalysts, jewellery, chemicals, hydrogen electrolysis	LPPM, NYMEX
Base metals	Copper, nickel, zinc, lead, aluminum, tin	Construction, power grids, vehicles, industrial production; strongly tied to China	LME (London), COMEX, SHFE (Shanghai)
Bulk and steelmaking	Iron ore, metallurgical coal, manganese	Steel output, above all in China	Index-priced: 62% Fe fines (SGX, Dalian); premium hard coking coal (Platts)
Battery and critical minerals	Lithium, cobalt, graphite, rare earths, vanadium	Electric vehicles, grid storage, permanent magnets, defence	Price assessments (Fastmarkets, Benchmark); GFEX lithium futures
Energy minerals	Uranium	Nuclear fuel procurement by utilities	Spot and long-term contract assessments (UxC, TradeTech)

The Value Chain

Where a company sits on the value chain sets how exposed its earnings are to the metal price.

The mining value chain runs from finding a deposit to delivering metal to a fabricator (Exhibit 1). Where a company sits on it sets how exposed its earnings are to the metal price.

Exhibit 1: Mining Value Chain



Source: Northwall Research

- **Upstream (explorers, developers, producers).** Owns the orebody and sells metal or concentrate at the market price less costs. Earnings move most with the commodity price: this is where the operating leverage sits, and where most listed mining equities live.

- **Midstream (smelters, refiners, traders).** Converts concentrate into refined metal for a processing fee (treatment and refining charges, TC/RCs) plus by-product sales. Margins depend on concentrate supply and demand more than on the metal price.
- **Downstream (fabricators and end users).** Turns refined metal into wire, rod, sheet, steel, batteries and jewellery. Metal is a pass-through cost.

Some companies choose one segment: a junior drilling a copper porphyry, a pure producer such as Freeport-McMoRan, or a royalty company such as Franco-Nevada that owns a slice of revenue without operating. Diversified majors such as BHP, Rio Tinto and Glencore span several commodities and, in Glencore's case, smelting and trading as well.

Geology & Deposit Types

An ore deposit is a rare, natural concentration of metal high enough to mine at a profit. Geology decides the three numbers that drive every mine's economics: size, grade and shape.

A mineable copper deposit needs 50 to 100 times the crustal average; a gold deposit 250 times or more.

The average crust holds about 0.006% copper and 0.004 g/t gold. A mineable copper deposit needs roughly 50 to 100 times that concentration, and a gold deposit 250 times or more. Geology decides size (tonnes), grade (metal per tonne) and shape (how deep, how continuous, how much waste must be moved).

How Ore Deposits Form

Most metal deposits form when hot, metal-bearing fluids move through rock and drop their metal where temperature, pressure or chemistry changes abruptly. Others form by settling in ancient seas or basins, by weathering that leaves metal behind (laterites), or by crystallizing in magma. The formation process controls a deposit's size, grade, depth and metallurgy, so geologists classify deposits into types with predictable characteristics. Exhibit 2 maps the type examples used in this primer.

Exhibit 2: Major Deposits of the World by Commodity



Source: Natural Earth (base map); company disclosures; Northwall Research

Major Deposit Types

Deposit type	Main metals	Typical size and grade	Mining style	Examples
Porphyry	Copper, gold, molybdenum	Very large (500 Mt to several Bt); low grade (0.3% to 1.0% Cu)	Large open pit, later block cave	Escondida (Chile), Grasberg (Indonesia), Oyu Tolgoi (Mongolia)
Sediment-hosted copper	Copper, cobalt	Large; higher grade (2% to 6% Cu)	Underground and open pit	Kamoa-Kakula (DRC), Zambian Copperbelt
Volcanogenic massive sulphide (VMS)	Copper, zinc, lead, silver, gold	Small to medium (5 to 100 Mt); high grade, polymetallic	Mostly underground	Kidd Creek (Canada), Neves-Corvo (Portugal)
Iron oxide copper-gold (IOCG)	Copper, gold, uranium	Large, variable grade	Underground	Olympic Dam (Australia)
Orogenic gold	Gold	Small to large; 2 to 10+ g/t in narrow veins	Underground, some open pit	Timmins and Kirkland Lake (Canada), Kalgoorlie (Australia)
Epithermal	Gold, silver	Small to medium; can be very high grade near surface	Open pit or underground	Lihir (Papua New Guinea), Yanacocha (Peru)
Banded iron formation	Iron ore	Billions of tonnes; 55% to 65% Fe	Massive open pit	Pilbara (Australia), Carajás (Brazil)
Magmatic sulphide	Nickel, copper, PGMs	Medium; 1% to 3% Ni or 3 to 10 g/t PGM	Underground	Sudbury (Canada), Norilsk (Russia), Bushveld (South Africa)
Nickel laterite	Nickel, cobalt	Large, shallow; 1% to 2% Ni	Shallow open pit, then HPAL or smelting	Sulawesi (Indonesia), New Caledonia
Lithium pegmatite and brine	Lithium	Hard rock 1% to 1.5% Li_2O; brine 500 to 2,000+ mg/L Li	Open pit (spodumene) or evaporation ponds	Greenbushes (Australia), Salar de Atacama (Chile)
Unconformity uranium	Uranium	Small footprint, extremely high grade (up to 15%+ U_3O_8)	Underground, freeze methods	Cigar Lake, McArthur River (Canada)
Sandstone uranium	Uranium	Low grade, permeable host rock	In-situ recovery (ISR)	Kazakhstan, Wyoming

Grade, Tonnage and Strip Ratio

Grade is the single most important number in mining. Continuity matters just as much.

- **Grade** is the concentration of metal: grams per tonne (g/t) for gold, silver and PGMs; percent for base metals and lithium oxide (Li_2O). A 2 g/t gold deposit produces twice the metal per tonne processed as a 1 g/t deposit for nearly the same mining cost.
- **Tonnage** is the mass of mineralized rock. Contained metal = tonnes × grade. Example: 100 Mt at 0.5% Cu contains 500,000 t of copper, or about 1.1 billion lb.
- **Cut-off grade** is the lowest grade that pays to process. Rising metal prices lower the cut-off and can grow reserves; rising costs do the opposite.
- **Strip ratio** is tonnes of waste moved per tonne of ore in an open pit. A 3:1 strip ratio means 3 t of waste for every 1 t of ore. Higher strip ratios raise unit costs and capital needs.
- **Metallurgy** is how easily the metal can be extracted. “Free-milling” gold leaches easily; “refractory” gold locked in sulphides needs roasting, autoclaves or bio-oxidation and costs more.

Geologists often note that grade is king, but continuity matters just as much. A high-grade deposit that is erratic and hard to model carries more risk than a lower-grade, predictable one.

Resources & Reserves

Mineral Reserves are the only part of a deposit a company can call economically mineable. Investors pay most for reserves, less for Measured and Indicated resources, and least for Inferred.

Under NI 43-101, a Qualified Person signs off every resource and reserve estimate.

Reporting Codes

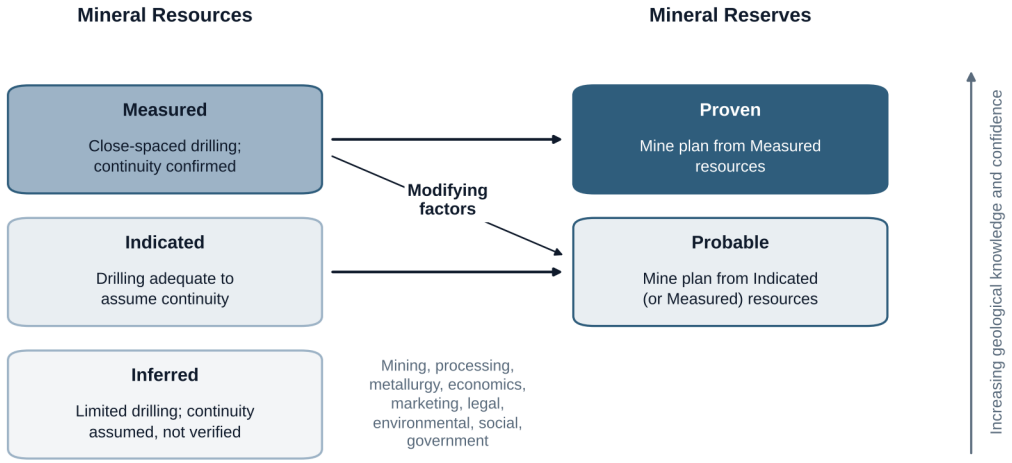
Public mining companies must report resources and reserves under a recognized code, signed off by a Qualified Person (QP) or Competent Person. The codes share one framework through CRIRSCO, the Committee for Mineral Reserves International Reporting Standards, so the categories line up across jurisdictions.

Code	Jurisdiction	Who signs off	Notes
NI 43-101 (with CIM Definition Standards)	Canada (TSX, TSX-V)	Qualified Person	Requires a technical report for material properties; the most common code for juniors worldwide
JORC Code	Australia (ASX); also Asia and Africa	Competent Person	Requires Table 1 disclosure of sampling and estimation methods
S-K 1300	United States (SEC registrants)	Qualified Person	Replaced Industry Guide 7 in 2021; now allows resource disclosure
PERC, SAMREC	Europe, South Africa	Competent Person	CRIRSCO-aligned

The Classification Ladder

Resources move up the ladder with more drilling, which raises geological confidence. They convert to reserves by applying **modifying factors**: mining method, dilution and recovery, metallurgy, costs, metal prices, infrastructure, permits, and environmental and social conditions (Exhibit 3).

Exhibit 3: Mineral Resource and Reserve Classification (CIM and CRIRSCO)



Source: CIM Definition Standards; CRIRSCO; Northwall Research

	Category	Confidence	Can it support a mine plan?
Reserves	Proven	Highest: converted from Measured resources with modifying factors applied	Yes: basis for feasibility studies and financing

	Category	Confidence	Can it support a mine plan?
	Probable	High: converted from Indicated (or Measured) resources	Yes
Resources	Measured	Close-spaced drilling; grade and continuity confirmed	Only after conversion to reserves
	Indicated	Drilling adequate to assume continuity	Only after conversion; usable in PFS/FS
	Inferred	Limited drilling; continuity assumed, not verified	No: only in PEAs, with cautionary language
Exploration target	Target or potential	Conceptual, no resource estimated	No: a range of tonnes and grade only

A Mineral Reserve is reported as the ore actually delivered to the mill, after dilution and mining losses, so reserve grade is normally lower than resource grade. Two reporting conventions deserve attention:

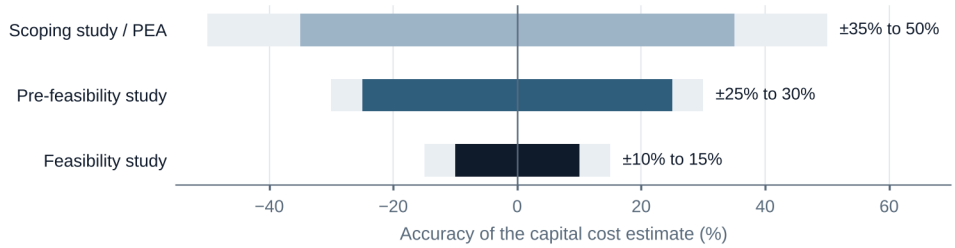
Adding resources to reserves when resources are reported inclusive double-counts metal.

- **Inclusive vs. exclusive.** Some companies report resources *inclusive* of reserves (the reserve is a subset); others report them *exclusive* (additional to reserves). Adding the two when resources are inclusive double-counts metal.
- **Price deck.** Reserves are calculated at a stated metal price. A company using US$2,000/oz gold for reserves while spot trades above US$4,000/oz has a conservative estimate with built-in upside; one using a price near spot has little cushion.

Study Stages

Each engineering study refines the estimate of capital, operating costs and economics, and costs more to complete (Exhibit 4).

Exhibit 4: Accuracy of the Capital Estimate by Study Stage



Source: CIM; AACE International; Northwall Research

Study	Resources it may use	Purpose
Scoping study / PEA (Preliminary Economic Assessment)	Measured, Indicated and Inferred	First look at economics; cannot declare reserves
Pre-feasibility study (PFS)	Measured and Indicated	Selects the preferred development option; first declaration of reserves
Feasibility study (FS or DFS)	Measured and Indicated	Basis for the construction decision and project finance

Operational Metrics

Persistently below 100% reserve replacement means the company is shrinking.

Reserve life (years) is reserves divided by annual production. A mine with 5 Moz of gold reserves producing 400 koz a year has a 12.5-year reserve life. Gold investors typically want 10+ years; shorter lives force a company to keep buying or discovering ounces.

Reserve replacement ratio is reserves added in a year divided by reserves mined. A producer that mined 500 koz and added 600 koz through drilling has a 120% replacement ratio. Persistently below 100% means the company is shrinking.

Conversion rate is the share of Inferred resources that become Indicated (or of resources that become reserves) after infill drilling. Historical conversion of 60% to 80% at a given mine is a useful guide for valuing its remaining Inferred ounces.

Discovery cost is exploration spend divided by new ounces or pounds discovered. A junior that spends US$20 million and adds 1 Moz of resources has a discovery cost of US$20/oz.

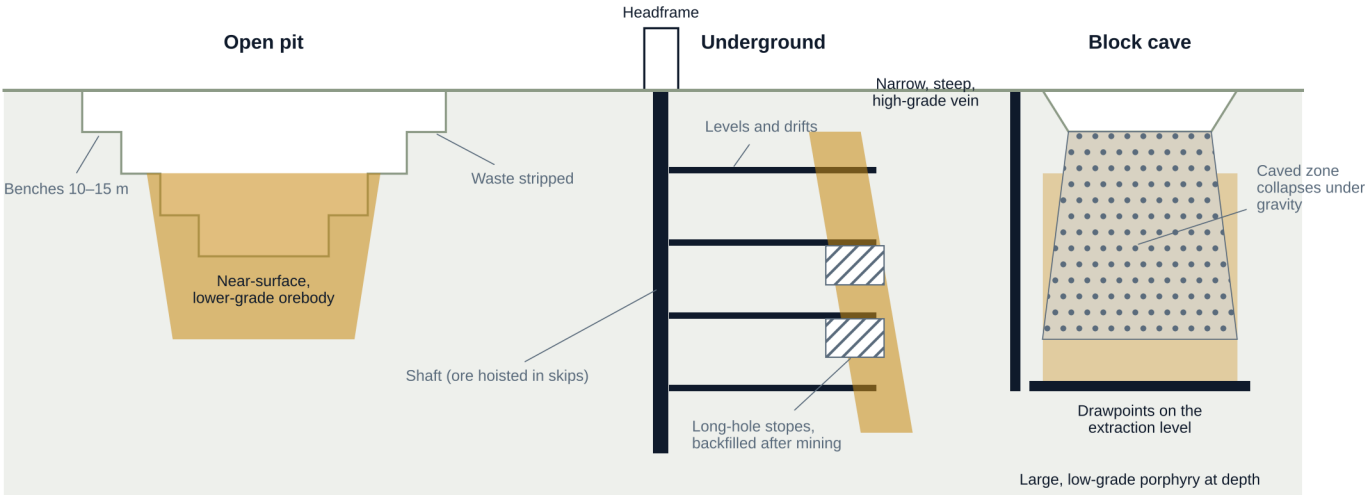
Mining Methods

The mining method is set by the orebody's depth, shape and grade, and it fixes most of the cost structure.

Open pits move huge tonnages cheaply but must strip waste; underground mines need grades several times higher.

Open pits move huge tonnages cheaply but must strip waste; underground mines move less rock at a higher cost per tonne, so they need higher grades (Exhibit 5).

Exhibit 5: Open Pit, Underground and Block Cave Mining (Schematic)



Source: Northwall Research; schematic, not to scale

Surface Mining

Open pit. Benches are drilled, blasted, loaded by shovels or excavators and hauled by trucks, widening and deepening the pit over time. This suits large, near-surface, lower-grade deposits (porphyry copper, bulk gold, iron ore). Mining costs are typically US$2 to US$5 per tonne of material moved, so economics hinge on the strip ratio and haul distances.

Strip mining. Used for flat-lying deposits such as coal, bauxite and mineral sands. Overburden is removed in strips, ore is mined, and waste is backfilled behind the advancing face.

Placer and alluvial mining. Recovers gold, tin, diamonds and mineral sands from loose sediments using gravity separation. Low capital, and often artisanal.

Underground Mining

Underground mines are accessed by a vertical **shaft**, which hoists ore in skips, or a **decline** (ramp), which trucks drive to surface. Costs typically run US$50 to US$150+ per tonne of ore, so grades must be several times higher than an open pit's.

Method	How it works	Suited to	Cost and recovery
Block caving	Undercut the base of the orebody; it collapses under gravity and is drawn from drawpoints below	Very large, low-grade bodies (porphyry copper at depth)	Lowest underground cost (US$10 to US$20/t); high capital, long build (5 to 10 years)
Sublevel caving	Ore is blasted level by level and the hanging wall caves behind	Large, steep bodies (iron ore, Kiruna)	Low cost; higher dilution
Long-hole (sublevel) open stoping	Large rooms blasted between levels, later backfilled	Steep, competent, medium-width bodies	Moderate cost; good recovery
Cut-and-fill	Ore mined in horizontal slices, each backfilled to form the floor for the next	Narrow, irregular or weak ground; high-grade veins	High cost; very selective, low dilution
Room-and-pillar	Rooms mined with pillars left for support	Flat-lying deposits (potash, some base metals)	Low cost; 20% to 40% of ore left in pillars

Solution Mining

In-situ recovery (ISR). A leaching solution is pumped through permeable ore via injection wells and recovered through production wells; no rock is moved. Most of Kazakhstan's and much of US uranium is produced this way, at low cost and with low surface disturbance.

Brine extraction. Lithium-rich brine is pumped from beneath salt flats into evaporation ponds that concentrate it over 12 to 24 months. Direct lithium extraction (DLE) uses adsorbents or ion exchange to cut this to days, but remains early in commercial scale-up.

Dilution and Mining Recovery

Grades persistently below reserve grade are one of the most common causes of guidance misses.

- **Dilution** is waste rock mined along with ore, lowering grade. 10% dilution turns 5.0 g/t in situ into roughly 4.5 g/t delivered to the mill.
- **Mining recovery** is the share of in-situ ore actually extracted; pillars, losses and unrecoverable edges reduce it.
- Both are built into reserves. When an operating mine reports grades persistently below the reserve grade, it is usually a sign that dilution or the geological model was too optimistic, one of the most common causes of guidance misses.

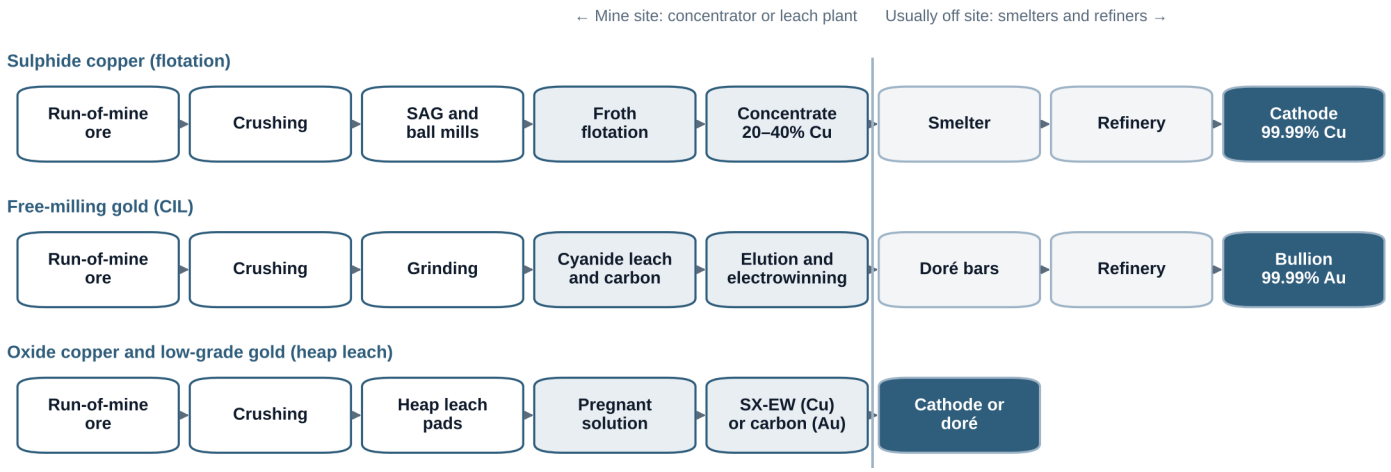
Processing & Metallurgy

Processing turns ore into a saleable product: a concentrate, doré bars or refined metal. Every point of recovery falls straight to revenue.

Recovery typically runs 85% to 95% for copper flotation and gold leaching.

Recovery, the share of metal in the ore that ends up in the product, typically runs 85% to 95% for copper flotation and gold leaching. Exhibit 6 shows the three most common routes from mine to metal.

Exhibit 6: Main Processing Routes, from Mine to Metal



Source: Northwall Research

Main Process Routes

Step	What happens	Used for
Comminution (crushing and grinding)	Ore is crushed, then ground in SAG and ball mills to sand or flour size to free the mineral grains	Almost all hard-rock ore; often the largest energy and cost item in the plant
Gravity separation	Dense particles separated by weight (jigs, spirals, centrifugal concentrators)	Coarse gold, tin, mineral sands, iron ore
Froth flotation	Chemicals make sulphide minerals water-repellent; air bubbles carry them into a froth skimmed off as concentrate (20% to 40% Cu, 50%+ Zn)	Copper, zinc, lead, nickel sulphides, PGMs
Cyanide leaching (CIL/CIP)	Ground ore is leached with dilute cyanide; dissolved gold is captured on activated carbon, stripped and poured as doré (gold-silver bars)	Free-milling gold
Heap leaching	Crushed or run-of-mine ore is stacked on lined pads and irrigated with cyanide (gold) or acid (copper)	Low-grade gold and oxide copper; low capital, lower recovery (50% to 80%)
Pressure oxidation, roasting, bio-oxidation	Breaks down sulphides that lock up gold before leaching	Refractory gold
SX-EW (solvent extraction, electrowinning)	Copper in acid leach solution is purified and plated directly as 99.99% cathode at the mine	Oxide copper; avoids smelting
HPAL (high-pressure acid leach)	Laterite ore leached with acid in autoclaves to produce nickel-cobalt intermediates	Indonesian nickel laterites
Pelletizing and beneficiation	Low-grade iron ore upgraded and formed into pellets	Magnetite iron ore
Spodumene concentration and conversion	Lithium ore concentrated to about 6% Li_2O (SC6), then refined into carbonate or hydroxide, mostly in China	Hard-rock lithium

Smelting and Refining

Most miners do not own smelters, so the terms on which concentrate is sold matter as much as the metal price.

Concentrate is sold to a smelter, which melts it to produce impure metal (blister or anode copper at about 99%), then refines it electrolytically to 99.99% cathode. China operates more than half of global copper smelting capacity, and most miners do not own smelters, so the terms on which concentrate is sold matter as much as the metal price.

How Concentrate Is Priced

A miner is not paid the full value of the metal in its concentrate. The smelter keeps a share and charges fees:

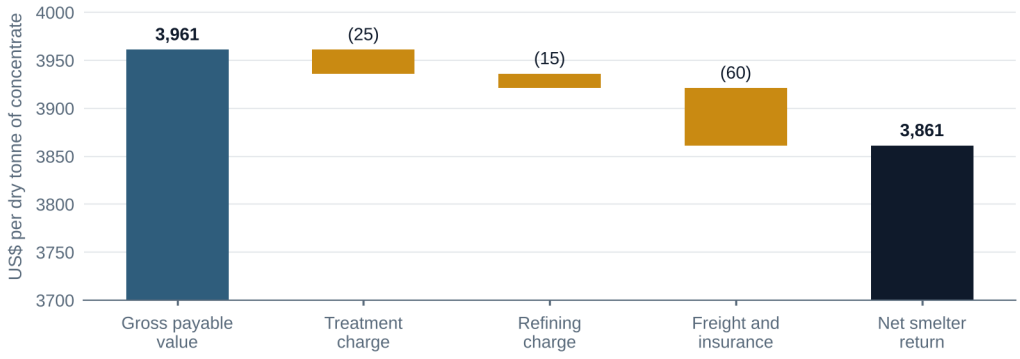
- **Payable metal.** Typically 96% to 97% of contained copper (or the contained amount less about one unit), 85% for zinc, and 90% to 97% of gold and silver in a copper concentrate.
- **Treatment charge (TC).** A fee per dry metric tonne of concentrate for smelting.
- **Refining charge (RC).** A fee per payable pound (copper) or ounce (precious metals) for refining.
- **Penalties.** Deductions for impurities above set limits, such as arsenic, bismuth, mercury or fluorine.
- **Freight and insurance** to the smelter port, plus the pricing period (often the average price in the month after arrival, called “M+1”).

TC/RCs are set annually in benchmark talks between major miners and Chinese smelters, and also trade on the spot market. When concentrate is scarce, as in 2025 when smelter capacity outgrew mine supply, spot TCs can fall to zero or below; smelters lose money while miners keep more of the metal value.

Worked example: net smelter return on 1 dry tonne of copper concentrate

Item	Calculation	US$ per dmt
Contained copper	28% grade × 2,204.6 lb = 617.3 lb	
Payable copper	96.5% × 617.3 lb = 595.7 lb	
Gross payable value	595.7 lb × US$6.65/lb	3,961
Less treatment charge	Illustrative US$25/dmt	(25)
Less refining charge	595.7 lb × US$0.025/lb	(15)
Less freight and insurance	Illustrative	(60)
Net smelter return		**3,861**

Exhibit 7: Net Smelter Return Bridge, One Dry Tonne of Copper Concentrate



Source: Northwall Research; illustrative terms

The miner keeps about 97% of the gross metal value at a US$25 TC. At an US$80 TC and 8¢ RC, levels seen in 2021 and 2022, the NSR falls by roughly US$90 per tonne, about 2.2%. For a mine selling 1 Mt of concentrate a year, that difference is about US$90 million of revenue.

The Commodities

Each metal has its own supply base, demand drivers and price mechanism. Copper and gold dominate mining-equity market value; iron ore drives the diversified majors' earnings.

Lithium, nickel and uranium are smaller markets but far more volatile.

Copper and gold dominate mining-equity market capitalization; iron ore drives the diversified majors' earnings; lithium, nickel and uranium are smaller but far more volatile.

Snapshot

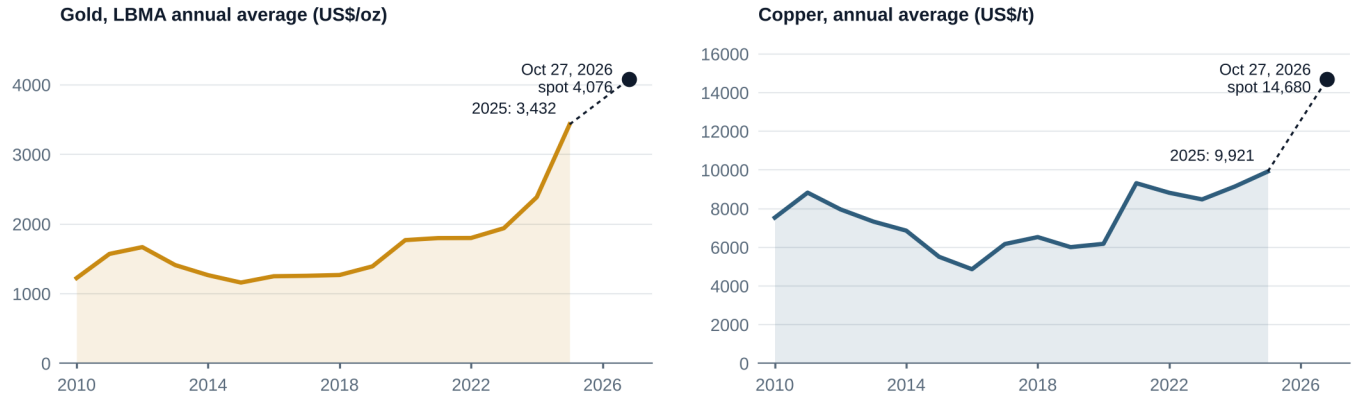
Commodity	Price quoted in	Main benchmark	Largest mining countries	Main demand
Copper	US$/t (LME) or US$/lb (COMEX)	LME Grade A cash	Chile, DR Congo, Peru	Power grids, buildings, vehicles, electronics, data centres
Gold	US$/oz	LBMA Gold Price, COMEX futures	China, Russia, Australia, Canada	Jewellery, bars and ETFs, central banks
Silver	US$/oz	LBMA Silver Price, COMEX	Mexico, China, Peru	Solar panels, electronics, jewellery, investment
Iron ore	US$ per dry tonne	62% Fe fines CFR China (SGX, Dalian)	Australia, Brazil	Steel
Metallurgical coal	US$/t	Premium hard coking coal FOB Australia	Australia, Canada, US, Mongolia	Steel (blast furnaces)
Nickel	US$/t	LME	Indonesia, Philippines	Stainless steel, batteries
Zinc and lead	US$/t	LME	China, Australia, Peru	Galvanized steel (zinc); lead-acid batteries (lead)
Aluminum	US$/t	LME plus regional premiums	China, India, Russia	Transport, packaging, construction, power lines
Lithium	CNY/t or US$/t of carbonate or hydroxide	GFEX futures, assessed prices	Australia, Chile, China, Argentina, Zimbabwe	EV and storage batteries
Uranium	US$/lb U_3O_8	Spot and long-term price assessments	Kazakhstan, Canada, Namibia	Nuclear power
PGMs	US$/oz	NYMEX, LPPM	South Africa, Russia	Autocatalysts, jewellery, industry

Where Prices Stand

Prices move daily: check a live quote before using any of these in a model.

Commodity	Price	Date and source
Gold	US$4,076/oz	October 27, 2026 close; Northwall market tape (Yahoo Finance, COMEX front month)
Silver	US$54.56/oz	October 27, 2026 close; Northwall market tape
Copper	US$6.66/lb (about US$14,680/t)	October 27, 2026 close, COMEX; LME cash US$14,751/t on October 14 (Farmonaut, citing Westmetall)
Platinum	US$1,543/oz	October 27, 2026 close; Northwall market tape
Palladium	US$1,034/oz	October 27, 2026 close; Northwall market tape
Nickel	About US$15,830/t	LME, October 2026 (Reporte Minero)
Aluminum	About US$3,231/t	LME, end of July 2026 (Mining Reporters)
Lithium carbonate	104,740 yuan/t (about US$15,650)	GFEX, October 12, 2026, roughly half its May 13 peak (Farmonaut, citing Reuters)

Exhibit 8: Gold and Copper Prices, Annual Averages 2010–2025 and Spot



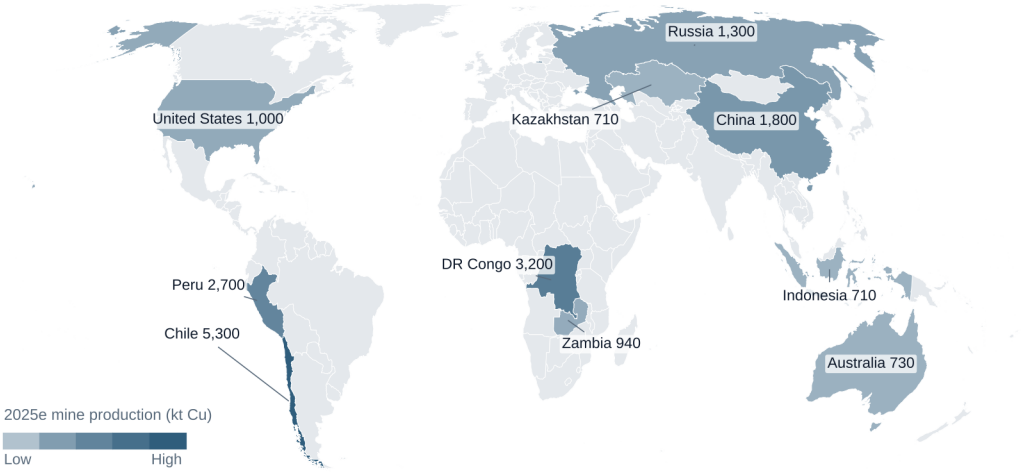
Source: LBMA (gold annual averages; 2025 per World Gold Council); LME (copper annual averages; 2025 from MetalCharts COMEX data); Northwall market tape (October 27, 2026 spot)

Copper

Copper is nicknamed "Dr. Copper" for its link to industrial activity.

Copper is the bellwether of the metals complex, nicknamed "Dr. Copper" for its link to industrial activity. About 23 Mt was mined in 2025, flat on 2024 despite strong demand, because of falling ore grades, mine closures and disruptions such as the September 2025 incident at Grasberg in Indonesia. China mines only about 8% of supply but refines roughly 14 Mt a year, more than half the world's refined copper.

Exhibit 9: Copper Mine Production by Country, 2025 Estimates (kt of Contained Copper)



Source: USGS Mineral Commodity Summaries 2026, via Critical Minerals News (world total about 23,000 kt); Natural Earth

- **Demand.** Roughly two-thirds of copper goes into electrical uses. Electrification (grids, EVs at three to four times the copper of a combustion car, renewables) and data-centre power are the growth themes on top of China's construction cycle.
- **Supply.** New mines take 15 years or more from discovery to production, average grades in Chile have fallen for two decades, and much new supply comes from higher-risk jurisdictions such as the DRC. This is why many analysts expect structural deficits.
- **By-products.** Gold, silver, molybdenum and cobalt credits can cut a copper mine's net cash cost substantially.

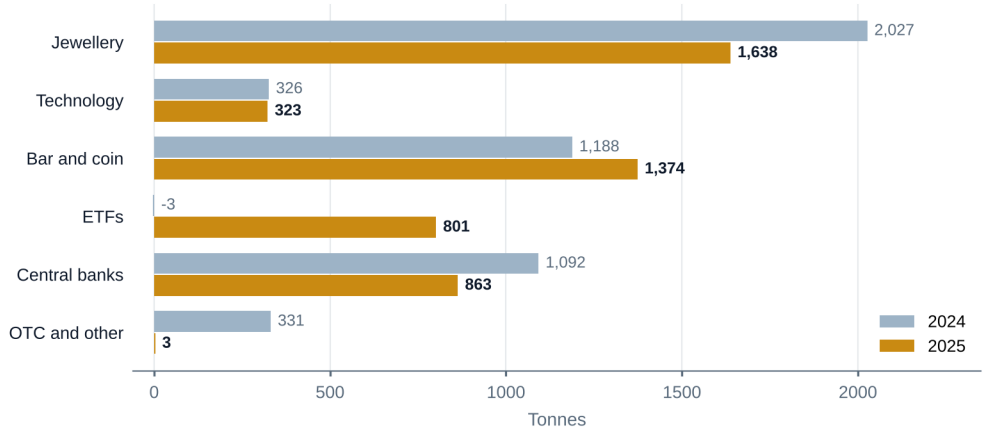
Gold

Above-ground gold stocks are about 60 times annual mine supply, so investment demand, not mine output, sets the price.

Gold is a monetary asset as much as a commodity. Above-ground stocks are about 60 times annual mine supply, so the price is set by investment demand rather than mine output. Mine production of about 3,670 t in 2025 (World Gold Council basis; USGS estimates about 3,300 t) has been broadly flat for a decade, topped up by about 1,400 t of recycling.

Total demand reached a record 5,002 t in 2025 (Exhibit 10). Investment demand almost doubled to 2,175 t as ETFs added 801 t, while jewellery fell 19% in tonnes as prices rose. The LBMA price averaged US$3,432/oz, up 44% on 2024, and gold passed US$5,000/oz for the first time in January 2026.

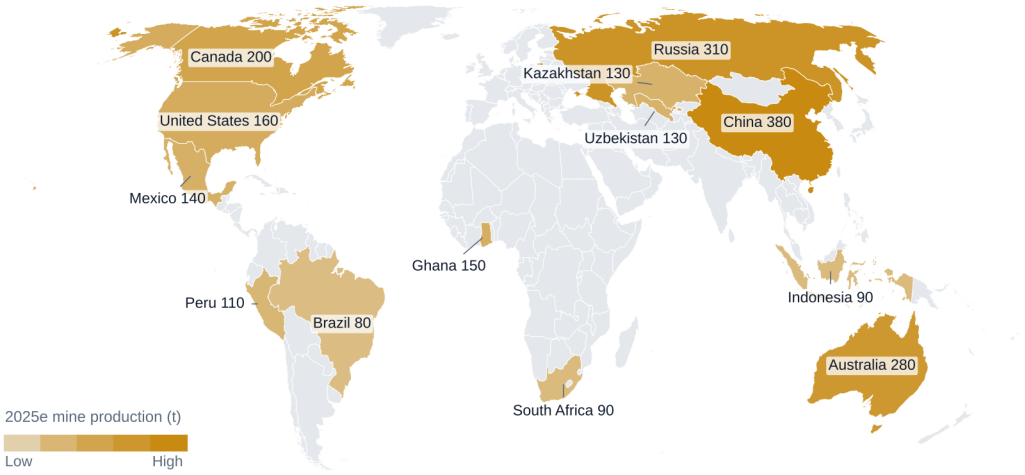
Exhibit 10: Gold Demand by Sector, 2024 vs. 2025 (Tonnes)



Source: World Gold Council, Gold Demand Trends Q4 and Full Year 2025 (Table 1)

Supply and demand (t)	2024	2025	Change
Mine production	3,650	3,672	+1%
Recycled gold	1,365	1,404	+3%
Net producer hedging	(54)	(74)	
Total supply	**4,962**	**5,002**	**+1%**
Jewellery fabrication	2,027	1,638	−19%
Technology	326	323	−1%
Bar and coin	1,188	1,374	+16%
ETFs and similar products	(3)	801	n/m
Central banks and other institutions	1,092	863	−21%
OTC and other	331	3	
Total demand	**4,962**	**5,002**	**+1%**
LBMA average price (US$/oz)	**2,386**	**3,432**	**+44%**

Exhibit 11: Gold Mine Production by Country, 2025 Estimates (Tonnes)



Source: USGS Mineral Commodity Summaries 2026 (world total about 3,300 t), via StatRanker; Natural Earth

- **Price drivers.** Real interest rates (gold pays no yield), the US dollar, central-bank buying (around 1,000 t a year from 2022 to 2024 and 863 t in 2025, led by emerging-market banks diversifying away from dollar reserves), ETF flows and geopolitical risk.
- **Equity leverage.** A miner with an AISC of US$1,800/oz earns a US$2,276/oz margin at US$4,076 gold. A 10% rise in the gold price lifts that margin by about 18%, which is why gold equities typically move 1.5 to 3 times the metal.

Silver

About 70% of silver is mined as a by-product of lead-zinc, copper and gold mines, so supply barely responds to price. Industrial uses, led by solar panels, now take over half of demand. Silver is far more volatile than gold: it set records above US$100/oz in late January 2026 (Host Rock weekly rundown) before falling back to about US$55/oz in October. Investors watch the gold-to-silver ratio, historically 50 to 80, as a relative-value signal; it stood near 75 at the end of October.

Iron Ore and Metallurgical Coal

China imports about 70% of seaborne iron ore, so Chinese steel output sets the price.

Iron ore is the largest mined commodity by tonnage and the biggest earnings driver for BHP, Rio Tinto, Vale and Fortescue. China imports about 70% of seaborne ore, so Chinese steel output, particularly for property construction, sets the price.

- **Pricing.** The 62% Fe fines index (CFR China) is the benchmark. Higher-grade ore (65% Fe, pellets) earns a premium because it cuts coke use and emissions; lower-grade and high-impurity ore (alumina, silica, phosphorus) trades at a discount.
- **Supply.** Australia's Pilbara and Brazil's Carajás dominate with very low costs (under US$25/t C1 for the best mines). Simandou in Guinea, led by Rio Tinto and Chinese partners, began its ramp-up in late 2025 and is the largest new source of high-grade supply.
- **Met coal.** Coking coal makes the coke that reduces iron ore in blast furnaces. Premium hard coking coal from Queensland is the price benchmark; Elk Valley Resources (formerly Teck's coal business, now majority-owned by Glencore) is the major Canadian supplier.

Nickel

Indonesia now supplies over half the world's nickel, using Chinese-built nickel pig iron plants and HPAL refineries to process laterite ore. The surge in low-cost supply drove prices down and forced

closures of higher-cost Western sulphide mines, including BHP's Nickel West in 2024. Investors distinguish Class 1 nickel (high purity, LME-deliverable, battery-grade) from Class 2 (ferronickel and NPI for stainless steel). Indonesian ore-mining quotas are now the single biggest swing factor for price.

Zinc and Lead

Zinc and lead usually occur together in the same deposits. Zinc's main use is galvanizing steel against corrosion; lead's is lead-acid batteries, much of which is recycled. Zinc miners are unusually sensitive to treatment charges because zinc concentrate pays only about 85% of contained metal.

Aluminum

Aluminum is made in three steps: bauxite is mined, refined into alumina (the Bayer process), then smelted into metal by electrolysis, which uses about 14 to 15 MWh of power per tonne. Power cost, not ore, sets the cost curve. China produces about 60% of global aluminum and has capped domestic capacity at about 45 Mt a year, which limits future supply growth.

Lithium and Battery Metals

Few metals show how fast high prices create their own supply.

Lithium is sold as spodumene concentrate (about 6% Li_2O, "SC6") from hard-rock mines, or as lithium carbonate and hydroxide from brines and converters. Prices are typically quoted per tonne of lithium carbonate equivalent (LCE).

- **Boom and bust.** Prices rose about tenfold in 2021 to 2022 on EV demand, collapsed by more than 80% by 2025 as Australian, African and Chinese supply surged, then recovered in 2026. Few metals show how fast high prices create their own supply.
- **Chemistry matters.** LFP (lithium iron phosphate) batteries, now dominant in China, use carbonate and no nickel or cobalt; high-nickel NMC batteries use hydroxide, nickel and cobalt.
- **Cobalt** is mostly a by-product of DRC copper mines; DRC export quotas have driven its price.
- **Graphite and rare earths** are dominated by Chinese processing, making them policy-sensitive "critical minerals" for Western governments.

Uranium

Uranium is sold mostly under long-term contracts between producers and nuclear utilities; the thinly traded spot market sets sentiment. Kazatomprom (Kazakhstan, mostly ISR) and Cameco (Canada, high-grade underground in the Athabasca Basin) are the two largest producers. Demand rests on reactor life extensions, new builds in China and India, and growing interest in small modular reactors and data-centre power. Supply has been slow to respond because new mines take a decade to permit and build.

Platinum Group Metals

South Africa's Bushveld Complex produces about 70% of the world's platinum; Russia's Nornickel is the largest palladium producer. Autocatalysts take most demand, so the shift to battery EVs is a long-term risk, partly offset by hybrids, jewellery and hydrogen technologies. Deep, labour-intensive South African mines are exposed to power outages and wage cycles. Platinum rose above US$2,700/oz in late January 2026 (Host Rock weekly rundown) and closed at US$1,543/oz on October 27.

Pricing & Markets

Most metals are priced off a handful of exchanges and assessment agencies. Over the long run, the marginal cost of supply anchors prices.

When price falls below the cost of the highest-cost producers needed to meet demand, supply is cut until the market rebalances.

A producer's realized price is the benchmark adjusted for product quality, location and contract terms. Over the long run, the marginal cost of supply anchors prices: when price falls below the cost of the highest-cost producers needed to meet demand, supply is cut until the market rebalances.

Exchanges and Benchmarks

Market	What trades there	Notes
London Metal Exchange (LME)	Copper, aluminum, zinc, nickel, lead, tin, cobalt	The global reference for base metals; cash and 3-month contracts; warehouse stocks watched as a supply signal
COMEX (CME Group, New York)	Gold, silver, copper	Copper quoted in US¢/lb; COMEX can trade at a premium to LME when US tariffs or import rushes distort flows
LBMA (London)	Gold and silver	Twice-daily gold auction sets the LBMA Gold Price used in most sales contracts and royalties
Shanghai Futures Exchange (SHFE)	Copper, aluminum, zinc, nickel, gold	Reflects Chinese domestic demand; the Shanghai-London arbitrage drives import flows
Guangzhou Futures Exchange (GFEX)	Lithium carbonate	Launched 2023; now the most-watched lithium price
SGX and Dalian Commodity Exchange	Iron ore	Futures settle against the 62% Fe index
Price reporting agencies (Platts, Argus, Fastmarkets, Benchmark, UxC)	Iron ore, coal, lithium, cobalt, rare earths, uranium	Assessed prices used where no liquid exchange contract exists

Realized Price vs. Benchmark

- **Quality premiums or discounts.** High-grade iron ore, low-impurity concentrate and battery-grade lithium all earn premiums; impurity penalties cut realized price.
- **Regional premiums.** Physical buyers pay a premium over the LME price for delivered metal: for example the Rotterdam or US Midwest aluminum premium, or the Shanghai Yangshan copper premium.
- **Provisional pricing.** Concentrate is invoiced provisionally on shipment and finalized months later at the average price of a quotational period. If prices fall between shipment and settlement, miners book negative "provisional price adjustments" in their next results.
- **Hedging and forward sales.** Locks in a price, protecting debt service but capping upside. Investors generally prefer unhedged gold producers; project lenders often require hedging.

Futures Curves: Contango and Backwardation

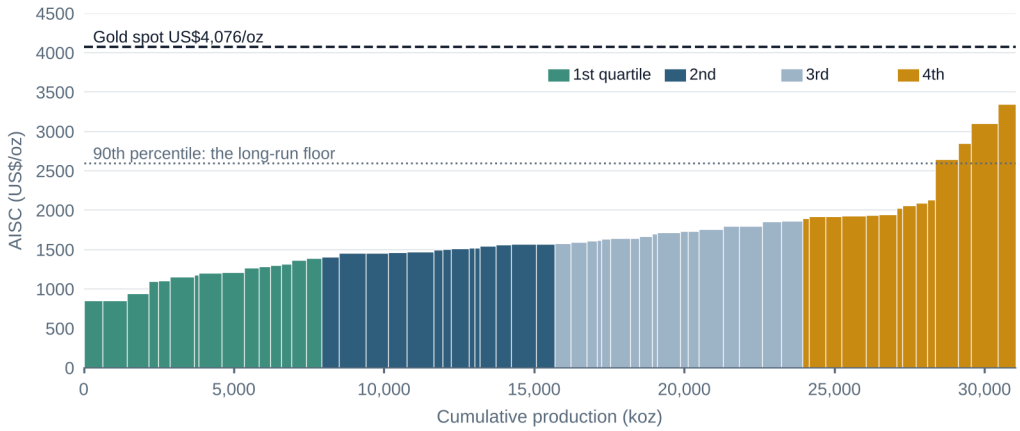
LME copper in backwardation with falling warehouse stocks is a classic bullish signal.

A futures curve is in **contango** when future prices are above spot, usually a sign of ample supply and inventory, and in **backwardation** when future prices are below spot, a sign of tight physical supply. LME copper in backwardation with falling warehouse stocks is a classic bullish signal. For investors holding futures or futures-based ETFs, contango creates a negative roll yield: rolling a US$10,000/t contract into a US$10,200/t contract each period loses 2% if the spot price stays flat.

The Cost Curve

An industry cost curve ranks every mine by unit cost (C1 or AISC) from lowest to highest, with cumulative production along the x-axis (Exhibit 12). Investors use it three ways:

Exhibit 12: Industry Cost Curve: How Investors Read It (Illustrative Gold AISC Curve)



Source: Northwall Research; illustrative distribution, not company data

1. **Price floor.** Prices rarely stay below the 90th-percentile cost for long: high-cost mines lose cash and close. That level is often treated as the long-term floor.
2. **Incentive price.** The price needed to justify building the next mine, including capital and a return, is usually well above today's cash costs. For copper it is generally estimated well above US$4/lb; long-term analyst price decks often sit near the incentive price.
3. **Position.** A mine in the first quartile survives any downturn; a fourth-quartile mine is a call option on higher prices.

Offtakes, Streams and Royalties

Developers fund mines partly by selling future production:

- **Offtake agreement.** A buyer (smelter, trader or end user) commits to purchase production at market-linked prices, often paired with a loan or equity investment.
- **Royalty.** A right to a percentage of revenue or profit from a mine (detailed under Fiscal Regimes), with no obligation to fund costs.
- **Stream.** An upfront cash deposit in exchange for the right to buy a fixed percentage of a mine's metal, often a by-product such as gold or silver from a copper mine, at a low fixed price (for example US$400/oz gold or 20% of spot). Wheaton Precious Metals, Franco-Nevada and Royal Gold are the leading streaming and royalty companies.

Streams and royalties are cheaper than equity for a developer when its share price is weak, but they reduce the cash flow left for shareholders and lenders, so they are deducted in the NAV.

Costs & Operating Metrics

The margin between price and unit cost is what investors buy. Learning to rebuild a mine's unit cost from its operating data is the core analytical skill.

Because costs are mostly fixed in the short run (labour, power, equipment), small changes in grade, throughput or price produce large changes in profit.

Cost Definitions

Measure	What it includes	Used for
Site or unit operating cost	Mining, processing and site G&A per tonne of ore (US$/t)	Comparing mines on an engineering basis
Cash cost (C1)	Site costs, plus TC/RCs, freight and royalties in some definitions, less by-product credits, per unit of payable metal	Base metals (copper, nickel, zinc); defines the industry cost curve
C3 (fully allocated)	C1 plus depreciation, corporate costs and interest	Accounting-based margin
All-in sustaining cost (AISC)	Cash costs, royalties and production taxes, sustaining capital, sustaining exploration, corporate G&A, reclamation accretion, less by-product credits (World Gold Council guidance, 2013)	Gold and silver; the main yardstick of free cash flow per ounce
All-in cost (AIC)	AISC plus growth capital and non-sustaining exploration	Total cost including expansion; whole-company cash burn

Cost measures are non-GAAP, so definitions vary between companies. Always check whether by-products are treated as credits (lowering cost) or co-products (costs split pro rata by revenue), and whether royalties, freight and TC/RCs are included.

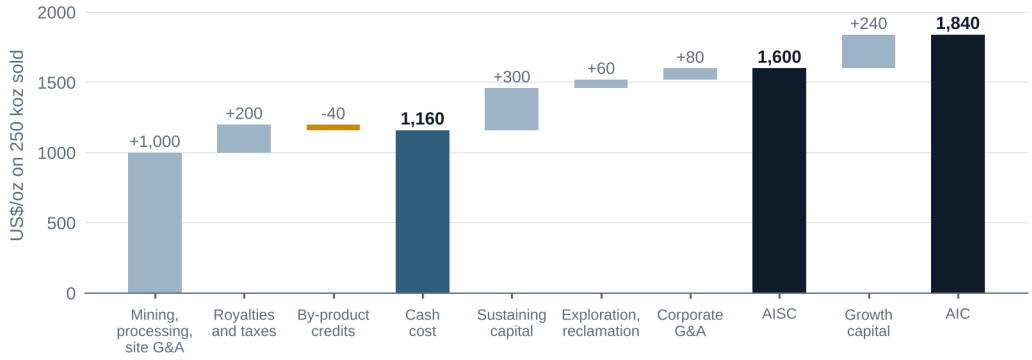
Key Operating Metrics

- **Throughput.** Tonnes of ore processed per day or year; capped by mill capacity.
- **Head grade.** The grade of ore actually fed to the mill. Compare it with reserve grade and with guidance each quarter.
- **Recovery.** The share of metal in the feed that reports to the product.
- **Production** = tonnes milled × head grade × recovery. Example: 10,000 tpd × 365 days × 1.5 g/t × 92% recovery = 5.04 t of gold, or about 162 koz a year.
- **Unit cost per tonne milled.** Divide total site costs by tonnes processed. A mine with US$60/t costs and 1.5 g/t ore at 92% recovery spends about US$1,350 per ounce produced before sustaining capital.
- **Sustaining vs. growth capital.** Sustaining capital keeps current output running (equipment replacement, tailings raises, underground development). Growth capital expands it. The classification matters because only sustaining capital is in AISC.

Worked Example: Gold AISC

Item	US$ million	US$/oz (250 koz sold)
Mining, processing, site G&A	250	1,000
Royalties and production taxes (5% of revenue at US$4,000/oz)	50	200
Less silver by-product credits	(10)	(40)
Cash cost	**290**	**1,160**
Sustaining capital	75	300
Sustaining exploration and reclamation	15	60
Corporate G&A	20	80
AISC	**400**	**1,600**
Growth capital and expansion exploration	60	240
AIC	**460**	**1,840**

Exhibit 13: From Site Costs to AISC and AIC (US$/oz)



Source: Northwall Research; worked example

At US$4,000/oz, this mine generates about US$2,400/oz of margin over AISC and US$600 million of pre-tax free cash flow before growth spending. Royalties rise with the gold price, so AISC is not fully fixed.

Worked Example: Copper C1 with By-Product Credits

A copper-gold mine produces 200 Mlb of payable copper and 100 koz of payable gold a year. Site costs are US$700 million, and TC/RCs plus freight are US$100 million.

- Gross cost: (US$700M + US$100M) / 200 Mlb = US$4.00/lb.
- Gold credit: 100 koz × US$4,000/oz = US$400 million, or US$2.00/lb.
- **C1 = US$2.00/lb** net of by-products.

By-product-heavy producers carry hidden exposure to the by-product price.

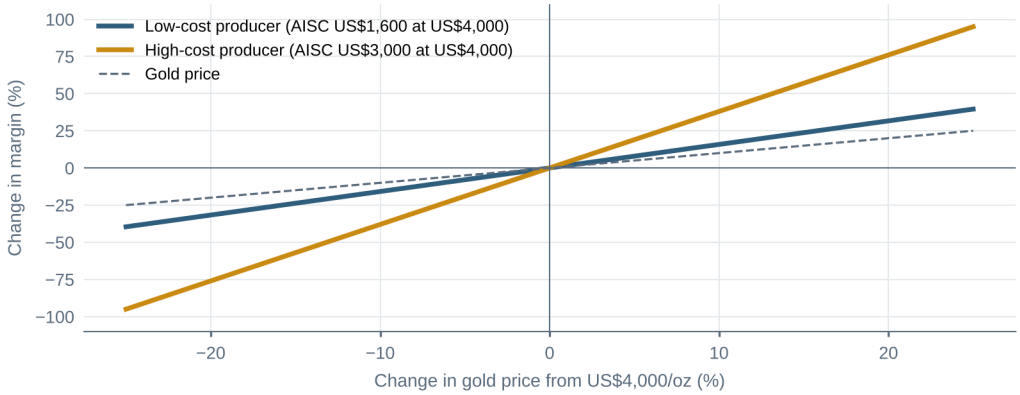
If gold falls 25% to US$3,000/oz, the credit falls to US$1.50/lb and C1 rises to US$2.50/lb, even though nothing changed at the mine. By-product-heavy producers carry hidden exposure to the by-product price.

Operating Leverage

Profit moves faster than price because costs are largely fixed. Using the gold example above (Exhibit 14):

Gold price (US$/oz)	Margin over AISC (US$/oz)	Change in margin vs. US$4,000
3,200	1,640	−32%
3,600	2,020	−16%
4,000	2,400	–
4,400	2,780	+16%
4,800	3,160	+32%

Exhibit 14: Margin Sensitivity to the Gold Price: Low-Cost vs. High-Cost Producer



Source: Northwall Research; royalties at 5% of revenue

The margin moves about 1.6 times the price (AISC is adjusted for royalties at 5% of revenue). Higher-cost producers have more leverage in both directions: at an AISC of US$3,000/oz, a 10% fall in gold from US$4,000 cuts the margin by about 38%.

Northwall coverage: Alamos Gold (NYSE/TSX: AGI)

Alamos shows why cost trajectory matters more than volume. Our model has production rising from 535 koz in 2026E to 770 koz in 2028E as the Island Gold Phase 3+ shaft and the Magino mill expansion arrive. Over the same period, AISC falls from about US$1,826/oz to US$1,350/oz.

At a flat US$4,000/oz gold price, the margin per ounce rises about 22%, but total margin rises about 75% because more ounces earn the wider margin. That combination of volume and falling cost is the core of our Buy rating and US$39.00 target.

Cost Inflation Drivers

Labour, energy (diesel and power are 15% to 30% of costs at many mines), reagents (cyanide, lime, sulphuric acid), explosives, tires and steel consumables, and local currency. Most mines report in US dollars but pay costs in Canadian or Australian dollars, Chilean pesos or South African rand, so a weaker local currency lowers US-dollar costs.

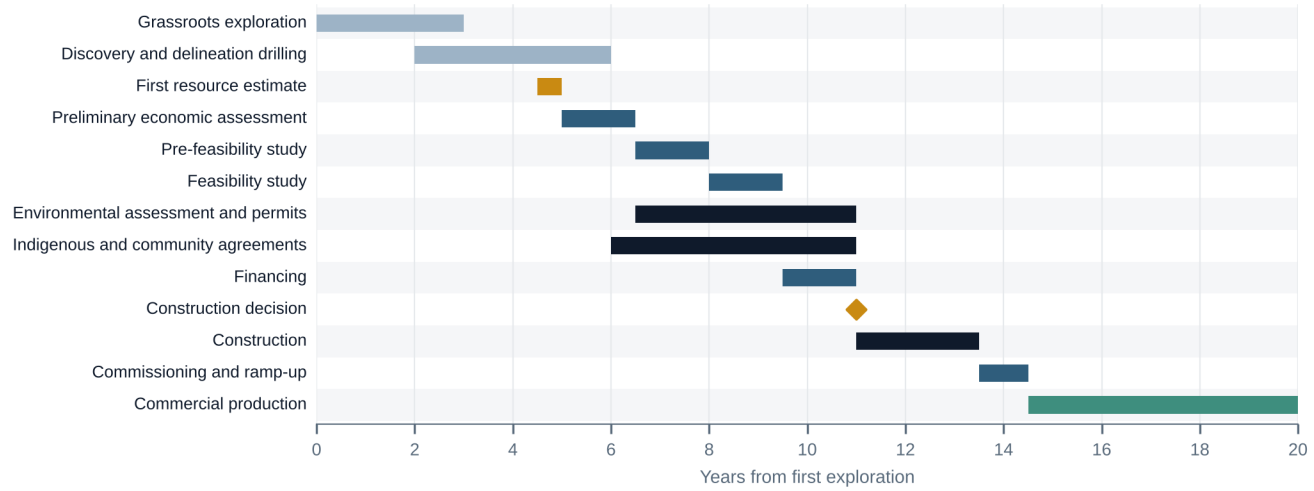
Project Life Cycle

A major new mine typically takes 10 to 20 years from discovery to first production, and only a small fraction of discoveries ever become mines.

Each stage removes a different risk, and a single-asset company's share price tends to follow a predictable path as those risks fall away.

Each stage removes a different risk, and the share price of a single-asset company tends to follow a predictable path as those risks fall away (Exhibit 15).

Exhibit 15: Typical Timeline for a Canadian Gold Project



Source: Company reports; Northwall Research; illustrative

Stages and Typical Durations

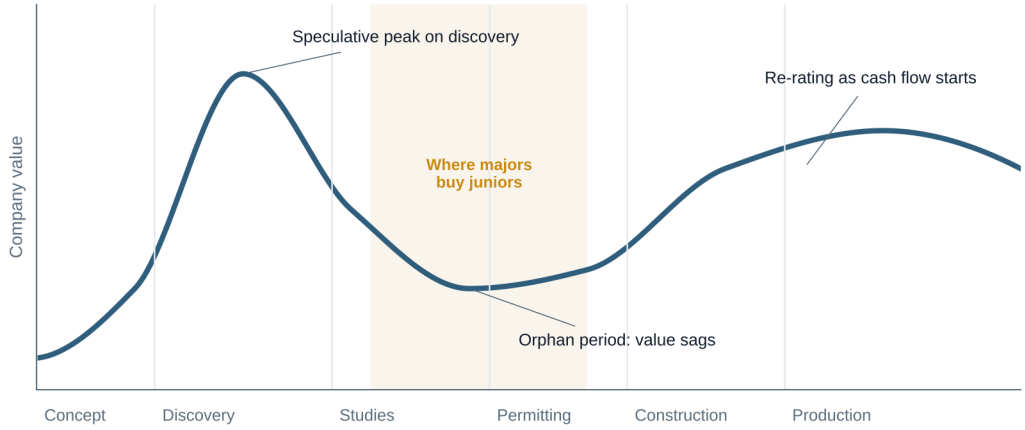
Stage	What happens	Typical duration	Main risk removed
Grassroots exploration	Mapping, geochemistry, geophysics, first drill holes	1 to 5 years	Is there mineralization?
Discovery and delineation	Step-out and infill drilling, first resource estimate	2 to 5 years	Is it big and continuous enough?
Studies (PEA, PFS, FS)	Metallurgical testwork, mine design, cost estimates, reserves	2 to 5 years	Is it economic?
Permitting and approvals	Environmental and social impact assessment, consultation, Indigenous agreements, permits	2 to 10 years	Can it be built?
Financing and construction	Debt, equity, streams; build mine, plant, tailings, power and roads	2 to 5 years	Can it be funded and built on budget?
Commissioning and ramp-up	Plant reaches design throughput and recovery	6 to 24 months	Does it work as designed?
Production	Steady-state mining, expansions, exploration near the mine	10 to 50+ years	Can it sustain output and costs?
Closure and reclamation	Decommissioning, land rehabilitation, water treatment, monitoring	Years to decades	Liabilities settled

The Lassonde Curve

The orphan period is where many acquisitions happen: majors buy de-risked deposits from juniors that cannot fund construction.

Named after Franco-Nevada co-founder Pierre Lassonde, the curve shows why junior mining stocks often peak on discovery headlines, fall through the “orphan period” of studies and permitting, and re-rate as construction is de-risked and cash flow begins (Exhibit 16). In the orphan period, speculators have left and producers’ investors have not yet arrived. It is where many acquisitions happen: majors buy de-risked deposits from juniors that cannot fund construction themselves.

Exhibit 16: The Lassonde Curve: Typical Share-Price Path from Idea to Mine



Source: Conceptual, after Pierre Lassonde; Northwall Research

Construction Risk

Capital cost overruns and delays are the norm rather than the exception: studies of large mining projects consistently find average overruns of 20% to 60%. Common causes are inflation in labour and materials, weaker ground or metallurgy than modelled, and scope changes. Ramp-ups that take longer than planned erode NPV because the most valuable early-year cash flows are pushed back.

ESG and the Social Licence to Operate

A permit is a legal licence; the social licence is the ongoing acceptance of communities, Indigenous rights-holders and governments.

A permit is a legal licence; the social licence is the ongoing acceptance of local communities, Indigenous rights-holders and governments. Losing it can stop a mine even after billions are spent. First Quantum's Cobre Panamá copper mine, one of the world's newest and largest, was ordered closed in late 2023 after nationwide protests and a Supreme Court ruling against its contract. Key ESG issues for mining investors:

- **Tailings.** Waste from processing is stored behind dams. The 2019 Brumadinho dam collapse in Brazil killed 270 people and led to the Global Industry Standard on Tailings Management (GISTM).
- **Water.** Use in arid regions (Chile, Peru, Australia), and acid rock drainage after closure.
- **Emissions.** Diesel haul fleets, grid power, smelting and, for steel, the coking coal and iron ore chain. Electrification and renewable power purchase agreements are now common.
- **Indigenous rights and community agreements.** Impact-benefit agreements, free, prior and informed consent, and local hiring.
- **Safety and labour.** Fatality rates, contractor safety, and artisanal mining in the supply chain (particularly DRC cobalt).

Closure and Reclamation

Mines must post closure plans and financial assurance (bonds, letters of credit) before construction. The present value of reclamation costs is a balance-sheet liability (asset retirement obligation) that analysts deduct in the NAV. Long-tail liabilities such as perpetual water treatment can make closure far more expensive than first estimated.

Fiscal Regimes & Jurisdictional Risk

Governments own most mineral rights and take their share through royalties, taxes and equity stakes. Jurisdiction is a direct input to valuation.

Total government take ranges from about 35% of pre-tax cash flow in mining-friendly jurisdictions to over 60% elsewhere.

Total government take typically ranges from about 35% of pre-tax cash flow in mining-friendly jurisdictions to over 60% where governments hold large free stakes or windfall taxes, and the rules can change mid-life. Jurisdiction therefore enters valuation directly, usually through a higher discount rate or a lower P/NAV multiple.

Royalty Types

Royalty	Basis	Who usually holds it	Comment
Net smelter return (NSR)	% of revenue after smelting, refining and transport deductions	Governments, royalty companies, prospectors	The most common private royalty; simple and paid from first production
Gross revenue royalty (GRR)	% of gross revenue, no deductions	Governments, royalty companies	Highest burden on the operator
Net profits interest (NPI)	% of profit after operating and often capital costs are recovered	Former owners, partners	Pays nothing until costs are recovered; harder to audit
Sliding-scale royalty	Rate rises with metal price	Governments, some private holders	Captures upside at high prices
Production-based (per tonne)	Fixed amount per tonne or ounce	Some governments	Simple; can bite hard at low prices

A 2% NSR on a mine with US$1 billion of annual net smelter revenue costs the operator US$20 million a year, roughly 5% of its value at typical margins. Royalty companies buy these interests from prospectors and juniors, building diversified portfolios with no operating cost exposure.

Other Fiscal Levers

- **Corporate income tax.** Commonly 25% to 35%, plus provincial or state taxes in Canada, Australia and the US.
- **Mining taxes.** Profit-based mining taxes (for example Ontario's 10% and Quebec's 16% to 28%) on top of corporate tax.
- **State participation.** Free-carried or paid equity stakes: the DRC's mining code requires a 10% free-carried interest; Indonesia required foreign-owned miners to divest 51% to Indonesian interests, which is how the state became majority owner of Grasberg.
- **Export duties and bans.** Indonesia's ban on unprocessed nickel ore exports (2020) built a domestic processing industry; Zambia and others have used concentrate export duties.
- **Windfall taxes and royalty reforms.** Chile introduced a new copper royalty for large producers in 2023; Mali rewrote its mining code in 2023 with higher state stakes.
- **Withholding taxes and repatriation limits** affect how easily cash can be returned to shareholders.

Resource Nationalism

Resource nationalism tends to rise late in commodity upcycles, exactly when miners' profits are largest.

When metal prices rise, governments often seek a larger share of the windfall through tax increases, contract renegotiations, forced local ownership or, at the extreme, expropriation. Recent examples include the closure of Cobre Panamá (2023), contract disputes and detentions of mine staff in Mali and Burkina Faso, the nationalization of lithium in Mexico (2022), and Chile's state-led

lithium strategy giving Codelco control of new Salar de Atacama projects. Resource nationalism tends to rise late in commodity upcycles, exactly when miners' profits are largest.

Assessing Jurisdictional Risk

Investors combine several inputs. The most widely cited is the Fraser Institute's Annual Survey of Mining Companies, which ranks jurisdictions on mineral potential and policy perception (Exhibit 17). In the 2025 survey, Nevada ranked first, with Ontario second and Saskatchewan third; China ranked last, behind Burkina Faso, Egypt, the Philippines and Mali.

Exhibit 17: Fraser Institute Investment Attractiveness Index, Top 10, 2025 Survey

Rank	Jurisdiction	Rank	Jurisdiction
1	**Nevada**	6	Western Australia
2	**Ontario** (15th in 2024)	7	Botswana
3	**Saskatchewan** (7th in 2024)	8	Norway
4	South Australia	9	Sweden
5	Arizona	10	Saudi Arabia

Source: Fraser Institute Annual Survey of Mining Companies 2025, via Mining Weekly (February 26, 2026)

- **Permitting timelines.** US federal permitting can take 7 to 10 years; Canadian timelines have also lengthened. Fast-track programs for critical minerals are shortening some.
- **Political and security risk.** Coups (Mali, Burkina Faso, Niger), conflict near operations, and sanctions (Russia).
- **Legal recourse.** Stabilization agreements and access to international arbitration offer some protection, but awards can take years to collect.

As a rule of thumb, sell-side analysts use 5% real discount rates for producing mines in top-tier jurisdictions and 7.5% to 10% for higher-risk countries, and apply lower P/NAV multiples to companies with assets in them.

Canada: A Top-Tier Mining Jurisdiction

Ontario ranked second in the world in the 2025 Fraser survey; Saskatchewan ranked third.

Canada combines world-class geology, deep capital markets on the TSX and TSX-V, and stable fiscal terms. Ontario and Quebec host the Abitibi greenstone belt, one of the most productive gold belts in the world; Saskatchewan's Athabasca Basin hosts the highest-grade uranium mines; and British Columbia's Golden Triangle and the Labrador Trough round out a diverse map (Exhibit 18).

Exhibit 18: Canada's Main Mining Camps and Northwall Coverage



Source: Natural Earth (boundaries); company disclosures; Northwall Research

Valuation

Mines are depleting assets with finite lives, so the sector's primary valuation method is net asset value, cross-checked against cash-flow multiples.

Net Asset Value (NAV)

Many analysts show NAV at a long-term deck, the futures strip and spot, because the answer changes dramatically.

NAV is the sum of discounted cash flows from each mine over its life, adjusted for corporate items. Stocks are then compared on price-to-NAV (P/NAV). For each asset, analysts build a life-of-mine model: annual tonnes mined and milled, grade, recovery, payable metal, price, operating costs, royalties, capital and taxes. The after-tax free cash flow is discounted to today.

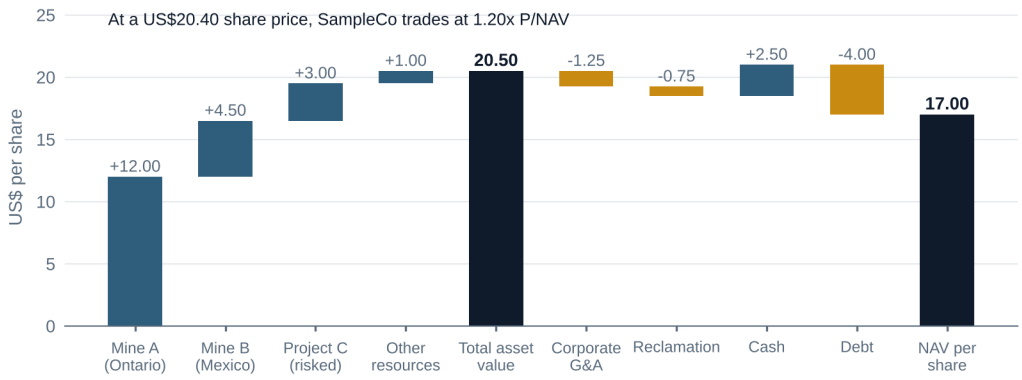
- **Discount rate.** Commonly 5% real for precious metals and 8% real for base metals and bulks, higher for riskier jurisdictions or development assets.
- **Price deck.** A long-term consensus price (typically well below spot in strong markets), the futures strip, or spot. Many analysts show NAV at all three, because the answer changes dramatically.
- **Corporate adjustments.** Add cash, investments and in-the-money options and warrants; subtract debt, the NPV of corporate G&A, reclamation liabilities, hedge losses and stream or royalty obligations.
- **Risking.** Development projects are multiplied by a probability factor that rises as they advance (for example 25% at PEA, 50% at PFS, 75% at a construction decision, 100% in production).

Resources outside the mine plan and exploration upside are added at a value per ounce or pound.

Example: SampleCo gold NAV (illustrative, US$4,000/oz long-term gold)

Asset	Stage	Discount rate	NPV unrisked (US$M)	Risk factor	NPV risked (US$M)	Per share (US$)
Mine A, Ontario	Producing	5%	2,400	100%	2,400	12.00
Mine B, Mexico	Producing	7.5%	900	100%	900	4.50
Project C, Nevada	Pre-feasibility	5%	1,200	50%	600	3.00
Resources outside mine plans	2 Moz at US$100/oz				200	1.00
Total asset value					**4,100**	**20.50**
Corporate G&A (NPV)					(250)	(1.25)
Reclamation obligations					(150)	(0.75)
Cash					500	2.50
Debt					(800)	(4.00)
Net asset value					**3,400**	**17.00**

Exhibit 19: SampleCo NAV Bridge (US$ per Share)



Source: Northwall Research; illustrative

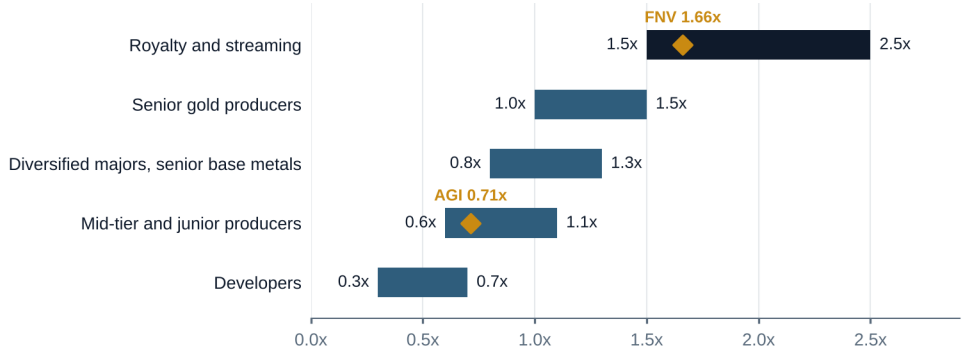
With 200 million shares and a share price of US$20.40, SampleCo trades at 1.20x P/NAV. If the market valued Project C at full NPV once it is built, NAV would rise by US$3.00 per share, a re-rating that can be anticipated as the project advances.

Price-to-NAV (P/NAV)

A stock trading well below peers on P/NAV may be cheap, or the market may doubt its reserves, costs, jurisdiction or capital allocation.

P/NAV captures what the market will pay for a dollar of discounted cash flow. Typical historical ranges vary through the cycle (Exhibit 20). A stock trading well below peers may be cheap, or the market may doubt the reserves, the cost guidance, the jurisdiction or management's capital allocation. The analyst's job is to find out which.

Exhibit 20: Typical P/NAV Ranges by Company Type, with Northwall Coverage



Source: Northwall Research; FNV and AGI at the October 26, 2026 close against Northwall NAV estimates

Company type	Typical P/NAV	Why
Royalty and streaming companies	1.5x to 2.5x	No cost exposure, free optionality on exploration, diversified
Senior gold producers	1.0x to 1.5x	Liquidity, long reserve lives, index inclusion
Diversified majors and senior base metals	0.8x to 1.3x	Long-life assets, cyclical earnings
Mid-tier and junior producers	0.6x to 1.1x	Fewer assets, higher operating risk
Developers	0.3x to 0.7x	Financing, permitting and construction risk
Explorers	Not meaningful	Valued on EV per resource unit or comparable discoveries

Trading Multiples

Multiple	Formula	Best for	What to watch
EV/EBITDA	(Market cap + net debt) / EBITDA	Majors and base-metal producers	Use forward EBITDA at a stated price deck; cyclical peaks flatter it
P/CF	Share price / operating cash flow per share	Gold producers	Cash flow before working-capital changes
Free-cash-flow yield	FCF / market cap	Mature producers returning capital	Sustaining vs. growth capex split
P/E	Share price / EPS	Majors and royalty companies	Distorted by impairments and depreciation
EV per ounce or pound in the ground	EV / attributable reserves and resources	Developers and explorers	Weight by category; adjust for grade, jurisdiction, metallurgy
EV per unit of annual production	EV / attributable annual output	Producer screening	Ignores reserve life and costs
Dividend yield and total shareholder return	Dividends and buybacks / market cap	Majors	Variable dividend policies tied to free cash flow

EV/EBITDA vs. P/CF. As with oil and gas producers, enterprise-value metrics are “debt neutral” and allow a cleaner comparison between companies with different balance sheets.

Valuing Royalty and Streaming Companies

Royalty companies are valued on P/NAV and P/CF, but at premiums to operators because they have no exposure to operating or capital cost inflation, carry little debt, and receive exploration upside on the land under their royalties for free. Key metrics are gold-equivalent ounces (GEOs)

sold, cash margin per GEO (often over 80%), portfolio diversification and the share of NAV from producing assets.

> **Northwall coverage: Franco-Nevada (NYSE/TSX: FNV)**
>
> Franco-Nevada earned a 91% adjusted EBITDA margin in H1 2026 with no asset above 10% of revenue and no debt. The market pays for that quality: at US$230.76 the shares trade at 1.66x our NAV per share of US$138.88, inside the 1.5x to 2.5x range for royalty companies.
>
> Our US$250.00 target blends 2.0x NAV with 25x 2027E cash flow per share. The 9.1% implied all-in return sits below our 15% Buy hurdle, so we rate the shares Hold: a premium franchise at a premium price.

Valuing Developers and Explorers

- **Developers** are valued on a risked NAV that rises as milestones are met (PFS, FS, permits, financing, construction). The usual question is what the stock is worth once built, discounted back for time and risk.
- **Explorers** have no cash flow. They are valued on EV per resource ounce or pound against comparable companies, adjusted for grade, depth, metallurgy and jurisdiction, or on the market's expectation of a discovery. Drill results, especially grade-thickness from step-out holes, drive the share price.
- **Takeover benchmarks.** Recent acquisition prices per ounce or per pound set reference values for similar deposits.

Investing: Commodities, Equities and Credit

Investors can own metal directly or indirectly through mining equities and bonds. Equities add operating leverage and growth, and with them operating, jurisdictional, cost and capital-allocation risk.

Direct Commodity Exposure

- **Physical metal and physically backed ETFs.** Common for gold, silver, platinum and, more recently, uranium (physical trusts). Return equals the price change less storage and management fees.
- **Futures.** Return = spot return + collateral return (interest on cash margin) + roll return. In contango the roll return is negative; in backwardation it is positive.
- **Commodity indices and broad ETFs.** Diversified, but base-metal weights are small and roll costs can erode returns.

Mining Equities

Tier	Typical profile	Examples	Investor appeal
Diversified majors	Multiple commodities and continents; iron ore, copper, coal, aluminum	BHP, Rio Tinto, Glencore, Vale, Anglo American	Dividends, balance-sheet strength, lower volatility
Senior producers	One main commodity, several mines, typically over 1 Moz gold or 500 kt copper a year	Newmont, Agnico Eagle, Barrick, Freeport-McMoRan, Southern Copper	Liquid leverage to one metal
Royalty and streaming	Portfolios of royalties and streams, no operations	Franco-Nevada, Wheaton Precious Metals, Royal Gold	Margin stability, optionality

Tier	Typical profile	Examples	Investor appeal
Mid-tier and junior producers	One to three mines	Many TSX- and ASX-listed names, including Alamos Gold	High leverage to price and execution
Developers	Building or financing a first mine		Re-rating as risk falls; financing dilution risk
Explorers	Drilling, no revenue	Thousands on the TSX-V and ASX	Discovery upside; most never become mines

Gold equities typically move 1.5 to 3 times the gold price in either direction. The correlation between a miner's shares and its metal varies with hedging, by-products, cost inflation, jurisdiction events and company-specific execution, and juniors often trade more on drill results and financing than on the metal price.

How Mines Are Financed

Source	Used by	Trade-off
Equity issues and private placements	Explorers, developers, all producers	Dilution; Canadian flow-through shares give investors tax deductions for exploration spending
Project finance debt	Developers with a feasibility study	Requires hedging, completion tests and cost-overrun facilities
Corporate bonds and revolving credit	Producers	Cheapest capital; covenants and refinancing risk
Streams and royalties	Developers and producers	No dilution, but a permanent claim on future revenue
Offtake-linked financing and strategic investors	Developers, especially battery metals	Often from traders, smelters, automakers or Chinese and Japanese buyers
Government and export-credit financing	Critical-minerals projects	Low cost but slow, with conditions on processing location and offtake

Mining Credit

Rating agencies stress-test credit metrics at their own conservative price decks, often well below spot.

Credit investors focus on a miner's ability to service debt through the bottom of the commodity cycle, not on upside. Growth spending, large acquisitions and shareholder returns funded by debt are generally viewed unfavourably by bondholders. Common credit metrics:

- Net debt / EBITDA (investment-grade miners usually target below 1.0x to 1.5x through the cycle)
- EBITDA / interest
- Funds from operations / debt
- Debt / proven and probable reserves (per ounce or pound)
- Reserve life relative to debt maturity
- Asset and commodity diversification, and position on the cost curve

Most majors are rated investment grade (A to BBB range); mid-tier producers and developers with project debt are typically high yield. Rating agencies stress-test credit metrics at their own conservative price decks, often well below spot.

Common Pitfalls

- **Extrapolating spot prices.** NAVs at spot can be 50% or more above NAVs at long-term prices in a strong market.

- **Ignoring dilution.** Share counts at juniors and developers often double before first production.
- **Trusting first-year guidance.** New mines frequently miss initial production and cost targets during ramp-up.
- **Overlooking taxes and royalties that rise with price,** which mute operating leverage.

Appendix I: Sample Drill-Results Press Release

The release below is fictional, written to show the language and data of a typical junior gold explorer's announcement. The company, project and figures are invented.

NORTHFOLD MINERALS CORP. INTERSECTS 18.4 METRES OF 6.2 G/T GOLD, EXTENDS RAVEN ZONE 300 METRES DOWN-PLUNGE

Fictional example · Timmins, Ontario

Northfold Minerals Corp. ("Northfold" or the "Company") is pleased to report results from 12 holes (6,850 metres) of its 2026 drill program at the 100%-owned Raven Gold Project, located 25 km from existing mills and highway access.

Highlights

- RV-26-041: **18.4 m of 6.2 g/t Au** from 412 m, including **4.1 m of 21.7 g/t Au**
- RV-26-044: 9.6 m of 4.8 g/t Au from 455 m, a 300 m step-out down-plunge of the current resource
- RV-26-038: 22.0 m of 1.9 g/t Au from 120 m, within the planned open-pit shell
- Mineralization remains open at depth and along strike

Hole	From (m)	To (m)	Length (m)	Au (g/t)
RV-26-041	412.0	430.4	18.4	6.2
including	418.2	422.3	4.1	21.7
RV-26-044	455.0	464.6	9.6	4.8
RV-26-038	120.0	142.0	22.0	1.9

Reported intervals are core lengths; true widths are estimated at 70% to 80% of core length. Assays are uncapped. The current Mineral Resource (2025) is 1.4 Moz Indicated at 2.1 g/t and 0.9 Moz Inferred at 2.6 g/t Au. Northfold plans an updated resource and a Preliminary Economic Assessment in the first half of 2027. The Company has C$18 million in cash. Technical information has been reviewed by a Qualified Person as defined by NI 43-101.

Key Take-Aways and Follow-Up Questions

"18.4 metres of 6.2 g/t gold"

Take-away. Grade × width (gram-metres) is the quick measure of a hole's quality: 18.4 × 6.2 = 114 g·m, well above the 30 to 50 g·m an underground deposit usually needs to be interesting. It is also well above the 2.1 g/t resource grade.

Follow-up. How much of the grade comes from the 4.1 m high-grade core? Without it, the remaining 14.3 m average about 1.8 g/t. Are assays capped in the resource model? High-grade gold is often "nuggety," and capping can cut resource grade by 10% to 30%.

"Extends Raven Zone 300 metres down-plunge"

Take-away. Step-out holes that extend mineralization grow the resource; infill holes inside it raise confidence but add few new ounces. A 300 m extension can add significant ounces if continuity holds.

Follow-up. Is one hole enough to infer continuity over 300 m? What is the drill spacing in the current Indicated resource? How deep is 455 m for an open pit versus an underground mine, and what does that do to costs?

“True widths estimated at 70% to 80% of core length”

Take-away. Core length overstates the true thickness of an inclined zone. At 75%, the 18.4 m interval is about 13.8 m true width, still a robust, mineable thickness.

Follow-up. Why are true widths estimated rather than known? Releases that omit true width deserve extra scrutiny.

“1.4 Moz Indicated at 2.1 g/t and 0.9 Moz Inferred at 2.6 g/t”

Take-away. About 2.3 Moz in total, 61% in the higher-confidence Indicated category. The Inferred portion cannot be used in a PFS until upgraded.

Follow-up. What gold price and cut-off grade underpin the resource? Is it constrained by a pit shell and underground stope shapes (“reasonable prospects”)? What are the metallurgical recoveries? A resource with 95% recovery is worth far more than one at 80%.

“25 km from existing mills and highway access”

Take-away. Proximity to infrastructure lowers capital and could allow ore to be toll-milled, or the project to be sold to a nearby producer. It makes Northfold a plausible takeover target.

Follow-up. Who owns the nearby mills, and do they have spare capacity? What is the permitting path and the status of agreements with local First Nations?

“C$18 million in cash”

Take-away. Funds the next 12 to 18 months of drilling and studies, but a PEA, then a PFS, will require more capital.

Follow-up. What is the burn rate? How many shares and warrants are outstanding, and at what prices could financing dilute existing holders?

Valuation check. If Northfold trades at an enterprise value of C$150 million, the market is paying about C$65 per resource ounce (C$150M / 2.3 Moz). Compare that with peers at a similar stage, grade and jurisdiction, and with recent takeover prices per ounce in the region.

Appendix II: Glossary of Terms

Acid rock drainage (ARD) Acidic, metal-laden water formed when sulphide minerals in waste rock or tailings oxidize; a major closure liability.

All-in cost (AIC) AISC plus growth capital and non-sustaining exploration.

All-in sustaining cost (AISC) Cash costs plus sustaining capital, exploration, corporate G&A and reclamation accretion, less by-product credits, per ounce sold.

Assay Laboratory analysis that measures the metal content of a rock sample.

Autoclave A pressure vessel used to oxidize refractory ores or leach laterites at high temperature and pressure.

Backfill Waste rock, tailings or cement paste placed in mined-out underground voids for support.

Ball mill Rotating cylinder with steel balls that grinds ore to a fine powder.

Bauxite The ore of aluminum.

Bench A horizontal step in an open pit, typically 10 to 15 m high.

Beneficiation Any process that upgrades ore before smelting or sale.

Blister copper Impure (about 98% to 99%) copper produced by a smelter, before refining.

Block cave Bulk underground method in which ore collapses under gravity after being undercut.

Brownfield Exploration or expansion at or near an existing mine.

By-product A metal recovered incidentally to the main product; its revenue is usually credited against costs.

C1 cash cost Direct cash cost per unit of payable metal, net of by-product credits.

Capping (top-cutting) Limiting extreme assay values in a resource estimate to avoid overstating grade.

Cathode Refined metal sheet (for copper, 99.99% pure) produced by electrolysis.

CIL / CIP Carbon-in-leach / carbon-in-pulp: gold recovery onto activated carbon during or after cyanide leaching.

Co-product A metal important enough that costs are allocated between it and the main product.

Collar The top of a drill hole.

Comminution Crushing and grinding of ore.

Competent Person / Qualified Person Professional who takes responsibility for resource and reserve estimates under JORC, NI 43-101 or S-K 1300.

Concentrate Upgraded product of a mill (for example 25% to 35% Cu), sold to a smelter.

Contained metal Tonnes × grade, before recovery losses.

Core Cylindrical rock sample from diamond drilling.

Cut-off grade Minimum grade at which material is classified as ore.

Cyanidation Leaching gold and silver with dilute cyanide solution.

Decline (ramp) Sloped tunnel providing vehicle access to an underground mine.

Diamond drilling Drilling with a diamond-studded bit that recovers solid core.

Dilution Waste mixed with ore during mining, lowering grade.

Doré Unrefined gold-silver bars poured at a mine site.

Drawpoint Opening from which caved ore is loaded in a block or sublevel cave.

Electrowinning (EW) Recovering metal from solution by electrolysis.

Epithermal Shallow, hot-spring-related gold and silver deposits.

Exploration target Conceptual estimate of a range of tonnes and grade, not a resource.

Feasibility study (FS) Detailed engineering study (±10% to 15% accuracy) used for construction and financing decisions.

Flotation Separation of sulphide minerals into a froth concentrate using reagents and air.

Flow-through shares Canadian shares that pass exploration tax deductions to investors.

Free-carried interest A stake (often held by a government) that does not contribute to capital costs.

Free-milling Ore from which gold is easily recovered by cyanidation.

Gangue Worthless minerals in ore.

Gold equivalent (AuEq) Multi-metal grade or production expressed as gold at stated prices and recoveries.

Grade Concentration of metal in rock (g/t or %).

Gram-metre (g·m) Grade × interval length; a quick measure of a drill intercept's quality.

Greenfield Exploration in a new area with no existing mine.

Hanging wall / footwall Rock above / below an inclined orebody.

Head grade Grade of ore fed to the mill.

Heap leach Leaching crushed ore stacked on lined pads.

HPAL High-pressure acid leach, used for nickel laterites.

In-situ recovery (ISR) Extracting metal by circulating solution through the orebody without moving rock.

Indicated resource Resource with enough drilling to assume geological and grade continuity.

Inferred resource Resource with limited drilling; continuity assumed but not verified.

Infill drilling Drilling between existing holes to increase confidence.

Intercept A drill-hole interval reported with its length and grade.

Lassonde curve Typical share-price path of a mining company from discovery to production.

Laterite Weathered, near-surface deposit enriched in nickel, cobalt or aluminum.

LBMA London Bullion Market Association; sets the benchmark gold and silver prices.

LCE Lithium carbonate equivalent.

Leach Dissolve metal from ore using a chemical solution.

Life of mine (LOM) Expected remaining years of production under the mine plan.

LME London Metal Exchange, the benchmark for base metals.

Measured resource Highest-confidence resource category, from close-spaced drilling.

Metallurgy The science of extracting metal from ore; also a deposit's processing characteristics.

Mill Processing plant; also the grinding equipment within it.

Modifying factors Mining, processing, economic, legal, environmental and social factors used to convert resources to reserves.

Net profits interest (NPI) Royalty paid on profit after costs are recovered.

Net smelter return (NSR) Revenue after smelting, refining and transport deductions; also a royalty on that revenue.

NI 43-101 Canadian disclosure standard for mineral projects.

Offtake Agreement to buy a mine's future production.

Open pit Surface mine excavated in benches.

Ore Material that can be mined and processed at a profit.

Orebody A continuous, mineable concentration of ore.

Orogenic gold Gold deposits formed in mountain-building belts, often in quartz veins.

Overburden Soil and rock covering an orebody.

Payable metal Share of contained metal in concentrate that the smelter pays for.

Pegmatite Coarse-grained igneous rock; the host of hard-rock lithium.

Porphyry Large, low-grade copper-gold-molybdenum deposit associated with intrusions.

Pre-feasibility study (PFS) Intermediate study (±25% accuracy) selecting the development option and declaring reserves.

Preliminary economic assessment (PEA) Early economic study that may include Inferred resources.

Probable reserve Economically mineable part of Indicated (and sometimes Measured) resources.

Proven reserve Economically mineable part of Measured resources.

Provisional pricing Invoicing concentrate at an estimated price, finalized after the quotational period.

Quotational period (QP) Period whose average price sets the final invoice (for example M+1).

Reclamation Rehabilitating mined land after closure.

Recovery Share of metal in the feed recovered to final product.

Refractory Ore that needs pre-treatment (roasting, pressure oxidation, bio-oxidation) before gold can be leached.

Reserve Economically mineable part of a resource, after modifying factors.

Resource Concentration with reasonable prospects for eventual economic extraction.

Royalty Right to a share of a mine's revenue or profit.

Run-of-mine (ROM) Ore as delivered from the mine, before crushing.

SAG mill Semi-autogenous grinding mill that uses ore plus some steel balls as the grinding medium.

Shaft Vertical opening used to hoist ore and move people and materials.

Smelter Plant that melts concentrate to produce impure metal.

Spodumene (SC6) Lithium mineral; SC6 is concentrate at about 6% Li_2O.

Step-out Drill hole testing for extensions beyond known mineralization.

Stope Underground excavation from which ore is mined.

Stream Upfront payment in exchange for the right to buy a share of production at a low fixed price.

Strip ratio Tonnes of waste moved per tonne of ore in an open pit.

Sustaining capital Capital needed to maintain current production.

SX-EW Solvent extraction and electrowinning; produces copper cathode at the mine.

Tailings Finely ground waste left after processing, stored in tailings facilities.

Throughput Tonnes of ore processed per period.

Toll milling Processing ore at another company's mill for a fee.

Treatment and refining charges (TC/RC) Fees paid by miners to smelters and refiners.

True width Actual thickness of a mineralized zone, versus drilled length.

U_3O_8 Triuranium octoxide, the traded form of uranium concentrate ("yellowcake").

VMS Volcanogenic massive sulphide deposit, rich in copper, zinc, lead, silver and gold.

Waste Rock below cut-off grade that must be moved to reach ore.

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Important Disclosures

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Companies under coverage. References to Alamos Gold Inc. (Buy, US$39.00 target) and Franco-Nevada Corporation (Hold, US$250.00 target) reflect Northwall Research's published ratings as of October 27, 2026. See the company reports for full methodology, risks and disclosures. Other companies are named for illustration only and are not covered.

Rating definitions. Buy: expected 12-month total return (price appreciation plus dividends) above 15%. Hold: expected total return between 0% and 15%. Sell: expected total return below 0%. Price targets are 12-month.

Sources. Facts are attributed to their sources in the text and the bibliography. Estimates, illustrations and opinions are Northwall Research's own. Market data are as at the dates stated and will change.

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